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Article *in* FEMS Microbiology Letters · July 2021

DOI: 10.1093/femsle/fnab090

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FEMS Microbiology Letters, 368, 2021, fnab090

https://doi.org/10.1093/femsle/fnab090
Advance Access Publication Date: 8 July 2021
Research Letter

RESEARCH LETTER – Physiology, Biotechnology & Genetics

A method for the enrichment, isolation and validation of *Mycobacterium smegmatis* population surviving in the presence of bactericidal concentrations of rifampicin and moxifloxacin

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One sentence summary: A method to enrich, isolate and validate rifampicin/moxifloxacin-surviving *Mycobacterium smegmatis* cells from the continuously antibiotic-exposed cultures for molecular studies.

Editor: Conor Meehan

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ABSTRACT

The bacterial populations surviving in the presence of antibiotics contain cells that have gained genetic resistance, phenotypic resistance and tolerance to antibiotics. Isolation of live bacterial population, surviving against antibiotics, from the milieu of high proportions of dead/damaged cells will facilitate the study of the cellular/molecular processes used by them for survival. Here we present a Percoll gradient centrifugation based method for the isolation of enriched population of *Mycobacterium smegmatis* surviving in the presence of bactericidal concentrations of rifampicin and moxifloxacin. From the time of harvest, throughout the enrichment and isolation processes, and up to the lysis of the cells for total RNA preparation, we maintained the cells in the presence of the antibiotic to avoid changes in their metabolic status. The total RNA extracted from the enriched population of live antibiotic-surviving population showed structural integrity and purity. We analysed the transcriptome profile of the antibiotic-surviving population and compared it with the orthologue genes of *Mycobacterium tuberculosis* that conferred antibiotic tolerance on tubercle bacilli isolated from the tuberculosis patients under treatment with four antitubercular antibiotics. Statistically significant comparability between the gene expression profiles of the antibiotic tolerance associated genes of *M. smegmatis* and *M. tuberculosis* validated the reliability/utility of the method.

Keywords: Mycobacteria; antibiotic-surviving population; rifampicin; moxifloxacin; gene expression profile; microarray

INTRODUCTION

Bacterial populations exposed to high doses of antibiotics often do not get completely eliminated. A small proportion of the population survives, and the antibiotic-surviving population consists of diverse phenotypes. These include antibiotic persisters (both non/slow-growing and non/slow-dividing), cells tolerant to a wide range of antibiotic concentrations, genetic

Received: 14 January 2021; **Accepted:** 6 July 2021

and phenotypic resisters, and many other phenotypes surviving by several mechanisms reported (Bigger 1944; Hui, Gordon and Kajioka 1977; Bentrup and Russell 2001; Kohanski, DePristo and Collins 2010; Sarathy, Dartois and Lee 2012; Wakamoto *et al.* 2013; Javid *et al.* 2014; Andersson and Hughes 2014; Brauner *et al.* 2016; Long *et al.* 2016; Van den Bergh, Fauvart and Michiels 2017; Hoeksema, Brul and Ter Kuile 2018; Ehrt, Schnappinger and Rhee 2018; Zhu *et al.* 2018; Balaban *et al.* 2019; Dewatcher, Fauvart and Michiels 2019; Nicoloff *et al.* 2019; Sebastian *et al.* 2020). We recently showed the high frequency *de novo* emergence of mutants of *Mycobacterium tuberculosis* (pathogen) and *Mycobacterium smegmatis* (saprophyte-cum-opportunistic pathogen), which were genetically resistant to rifampicin or moxifloxacin and other antibiotics, from their *in vitro* cultures exposed continuously to bactericidal concentrations of these antibiotics (Swaminath et al. 2020a,b; Sebastian *et al.* 2017).

To study the mechanisms by which bacterial cells survive in the presence of antibiotics, it is necessary to isolate the live surviving population from the midst of a large pool of dead/damaged cells. One of the available methods to isolate antibiotic-surviving bacterial cells is based on the lysis of 'non-persistent' cells using high doses of antibiotics followed by the sedimentation of the surviving population, as shown in the case of *M. tuberculosis, Escherichia coli*, *Pseudomonas fluorescens* and *Staphylococcus aureus* (Keren *et al.* 2004; Keren *et al.* 2011; Kwan *et al.* 2013; Canas-Duarte, Restrepo and Pedraza 2014). Another method involved differential fluorescence-based sorting of cells between metabolically active (fluorescence positive) and inactive (fluorescence negative or low fluorescence) persister cells (Shah *et al.* 2006; Henry and Brynildsen 2016; Srinivas *et al.* 2020). Here, the sorting and processing of the cells in the absence of the antibiotic would have relieved the cells from the antibiotic stress, causing changes in gene expression. In another method, antibiotics-treated *E. coli* cells were collected by centrifugation and transferred to a fresh growth medium for short-term re-exposure to the same antibiotics (Kwan *et al.* 2013). Here also, gene expression profile would have changed during the transfer of the antibiotic-survivors into the fresh growth medium. A method using dielectrophoresis, performed in the absence of antibiotics, yielded 3×10^4 surviving cells from 10^7 cells/ml of *M. smegmatis* cells exposed to isoniazid for 24 hrs (Elitas *et al.* 2014). Here also, the absence of antibiotics during dielectrophoresis would have changed the gene expression pattern.

Bacterial populations, which were continuously exposed to antibiotics, would contain more proportion of damaged/dead cells than undamaged/live cells. The total RNA extracted from such a mixture would be degraded and hence not useful for molecular studies using gene expression analysis. Thus, it was necessary to develop a method by which the antibiotic-surviving population could be isolated in the continuous presence of the antibiotics so that gene expression in response to the antibiotic stress would remain unaltered. Here we present a Percoll gradient centrifugation based method, where all the steps right from the culture up to the lysis of the rifampicin/moxifloxacin-surviving *M. smegmatis* cells for total RNA extraction were performed in the continuous presence of the respective antibiotic. The reliability and utility of the method were validated by isolating total RNA of high integrity and purity and finding out the gene expression profile of the antibiotic-surviving cells.

MATERIALS AND METHODS

Bacterial cultures

Mycobacterium smegmatis mc^{2}155 (*Msm*) (Snapper *et al.* 1990) was cultured in Middlebrook 7H9 medium containing 0.2% glycerol and 0.05% Tween 80, at 37°C and 170 rpm. The antibiotics were added when the culture reached 0.6 $OD_{600\ nm}$ (mid-log phase). The CFUs were determined on Middlebrook 7H10 agar plates.

Exposure of *Msm* cultures to Rifampicin and Moxifloxacin

Msm mid-log phase cultures were exposed to rifampicin (1.75x MBC RIF) and moxifloxacin (3.75x MBC MXF), as described (Swaminath *et al.* 2020a,b). The 0 hr of the exposure was the antibiotic addition time. Among the four RIF/MXF-exposed biological replicate cultures, three cultures each were harvested at the 36-hr for RIF and 48-hr for MXF exposure for RNA extraction. The fourth culture was used for CFU determination every 12 hr, from 0 hr to 96 hr, for growth profile in the presence of RIF/MXF, as in our earlier studies (Swaminath *et al.* 2020a,b). There was no appreciable change in the RIF/MXF MBC in the medium during the 96-hr cultures in our earlier studies (Swaminath *et al.* 2020a,b). In any case, rifampicin half-life is 7 days in the 7H9 medium (Yu *et al.* 2011). Hence, it was not determined again here.

Percoll gradient fractionation of the cultures

The cells harvested from the RIF- and MXF-exposed cultures were fractionated by Percoll gradient centrifugation in SW41 rotor in Beckman L8-70M ultracentrifuge (Beckman Coulter), at 385 x g for 1 hr, at 20°C, as described (Vijay *et al.* 2017). Details are given under Supplementary Materials and Methods.

SYTO9/propidium iodide staining for live/dead cell count

The cells from each fraction of the Percoll gradient were stained with SYTO9/propidium iodide (PI) (LIVE/DEAD *Bac*Light Bacterial viability kit, Invitrogen) and examined under DIC and fluorescence. Green (SYTO9 positive), yellow (SYTO9 and PI stained) and red (PI stained) cells were counted using the multipoint tool of ImageJ 1.51 j8. The percentage of cells with green, yellow, and red fluorescence were calculated, and graphs were constructed using MS-Excel. Details are given under Supplementary Materials and Methods.

Total RNA extraction, microarray and data analysis

Total RNA was extracted, as reported (Wecker 1959; described in detail in the Supplementary Materials and Methods), with minor modifications. To ensure the absence of DNA contamination, RNA samples were treated with DNase I (Thermo Scientific) and confirmed using PCR with 16S rDNA primers [Msm-16S-RT-f (5′ gcggtaatacgtagggtccg 3′) and Msm-16S-RT-r (5′ ttccagtctcccct-gcagta 3′)]. The 23S/16S rRNA ratio and the RNA integrity number (RIN) were determined using bioanalyser (Genotypic Technology Pvt Ltd, Bangalore, India). The detailed procedure of microarray is presented in the Supplementary Materials and Methods.

RESULTS

Experimental rationale/strategy

Mycobacterium smegmatis (*Msm*) antibiotic-surviving population was generated by exposing the mid-log phase cultures to bactericidal concentrations of rifampicin or moxifloxacin for 96 hrs, as reported by us (Swaminath *et al.* 2020a,b). Percoll gradient centrifugation, performed as described (Vijay *et al.* 2017), was used for the enrichment of live cells surviving in the presence of rifampicin or moxifloxacin. We verified and confirmed that the extent of enrichment of the antibiotic-surviving population was comparable whether rifampicin/moxifloxacin was present or absent throughout the entire procedure of the isolation and enrichment. Nevertheless, for maintaining the gene expression status of the cells unaltered, as it were in the culture in the presence of the antibiotic, the Percoll gradient centrifugation and all the procedures until the lysis of the cells for total RNA extraction were performed in the presence of the same MBC of the antibiotic used in the culture. The cells from different Percoll fractions, after washing away the antibiotic, were sampled for CFU determination and for staining with SYTO9 and propidium iodide (PI). This gave the proportions of the CFU-positive, SYTO9-positive, and PI-positive cells.

Since SYTO9 is cell-permeable, it stains DNA & RNA in both the intact and damaged cells. On the contrary, since live cells are impermeable to PI, the PI stain can bind nucleic acids only in a damaged/dead cell. Due to the higher affinity of PI for nucleic acids, in a dead cell, PI displaces SYTO9 bound to nucleic acids and gives red fluorescence (Stocks 2004). The cells with yellow fluorescence, due to the retention of both SYTO9 (green) and PI (red), were considered as damaged cells (Roszak and Colwell 1987; Stiefel *et al.* 2015). Therefore, even though the cells with yellow fluorescence might have been live, they were not considered for the enrichment and extraction of total RNA.

To avoid these possibilities, along with scoring for the SYTO9-positive cells, we also determined the CFU of all the Percoll fractions. Thus, the Percoll fraction, which contained the highest proportion of CFU-cum-SYTO9-positive cells, was used for the isolation of the cells for RNA extraction. This fraction, which would have the maximum enrichment of the CFU-positive antibiotic survivors, would possibly have several of the different types of antibiotic-surviving cells as well. These may include classical and other types of persisters, genetic resisters, transient phenotypic resisters, antibiotic-tolerant cells, and probably cells having many other phenotypes (see the first paragraph under Introduction). Thus, the present method was designed for the maximal enrichment and isolation of the antibiotic-surviving population as a whole and not of any specific type of antibiotic survivor(s).

The integrity and purity of the total RNA, extracted from the CFU-cum-SYTO9-positive enriched fraction of antibiotic-surviving cells, was taken as an indicator of the absence of dead/damaged cells in the preparation. We determined the transcriptome profiles of the *Msm* rifampicin/moxifloxacin survivors and compared them with that of tubercle bacilli isolated from the tuberculosis patients treated for 14 days with isoniazid, rifampicin, pyrazinamide, and ethambutol (Walter *et al.* 2015). The statistical significance of the comparison was used to assess the reliability and utility of the method.

Msm population surviving in the presence of rifampicin

The *Msm* mid-log phase cultures, upon exposure to 75 $\mu g\ mL^{-1}$ of rifampicin (1.75x MBC, as determined by Swaminath *et al.* 2020b; the MIC for rifampicin was 50 $\mu g\ mL^{-1}$, Fig. S1, Supporting Information) for 96 hrs, showed three distinct phases of susceptibility as revealed by their CFU on antibiotic-free plates. The killing phase, which lasted up to 24 hr of exposure with a 4-$\log_{10}$ reduction in the CFU, was followed by the rifampicin-surviving phase (RSP) lasting up to 48 hr without an apparent change in the CFU (Fig. 1A). The RSP was followed by a regrowth phase with a steady increase in the CFU from 48 hr onwards up to the 96 hrs monitored (Fig. 1A). The culture from the 36 hr of rifampicin exposure, which was the mid-point of the RSP, was used for the enrichment of live cells using Percoll gradient centrifugation. Among the four cultures for which the CFUs were determined, three cultures were harvested at 36 hr. The fourth culture was allowed to proceed up to 96 hrs in the presence of rifampicin to establish the complete profile of its response to the antibiotic (Fig. 1A), as reported by us earlier (Swaminath *et al.* 2020b).

Msm population surviving against MBC of moxifloxacin

The *Msm* mid-log phase cultures, upon exposure to 0.5 $\mu g\ mL^{-1}$ (3.75x MBC, as determined by Swaminath *et al.* 2020a; the MIC for moxifloxacin was 0.0625 $\mu g\ mL^{-1}$, Fig. S2, Supporting Information) of moxifloxacin for 96 hrs, also showed a triphasic response in terms of CFU. The killing phase lasted up to 36 hrs of exposure with 5-$\log_{10}$ unit reduction in the CFU. This was followed by the moxifloxacin-surviving phase (MSP) up to 60 hr, which was ensued by the regrowth phase up to 96 hr (Fig. 1B). The cultures from the 48 hr of moxifloxacin exposure, which was the mid-point of MSP, were used for the enrichment of live cells. Here also, three cultures were harvested at 48 hr and the fourth culture was allowed to go up to 96 hr for determining growth curve (Fig. 1B), which was found to be similar to the triphasic growth curve reported by us for *Msm* cells against moxifloxacin (Swaminath *et al.* 2020a).

Live/Dead bacteria in the rifampicin/moxifloxacin-exposed cultures

The proportions of SYTO9- and PI-stained cells from the cultures of mid-log phase, 36 hrs of rifampicin exposure, and 48 hrs of moxifloxacin exposure were quantitated. The *Msm* mid-log phase cells showed 81% ± 4.68% of SYTO9-positive cells, 6% ± 4.28% of PI-positive cells, and 12% ± 3.55% cells having both SYTO9 and PI (yellow fluorescence) (Fig. 2A and D). The 36-hr rifampicin-surviving population showed only 24% ± 4.5% of SYTO9-positive cells (~4-times less than that in the mid-log phase population), 52% ± 13.8% of PI-positive cells, and 24% ± 9.4% of yellow fluorescing cells (Fig. 2B and E). The 48-hr moxifloxacin-surviving population showed 44% ± 2.36% of SYTO9-positive cells, 27% ± 4.46% of PI-positive cells, and 27% ± 2.36% of yellow fluorescing cells (Fig. 2C and F).

Enrichment of CFU-cum-SYTO9-positive cells from the rifampicin-survivors

Different ranges of Percoll gradients were set up to get CFU-cum-SYTO9-positive enriched population from the 36 hr rifampicin-surviving cultures (Supplementary Text and Figs. S3–S7 in the Supplementary Materials). The 48%–68% range of the Percoll gradient, in the absence of rifampicin, with 4% increment of Percoll, gave the maximum possible enrichment (83.38%) of SYTO9-positive cells without any PI-positive cells (Fig. S5, Supporting Information). The same 48%–68% Percoll gradient centrifugation, when performed in the presence of 1.75x MBC rifampicin, also showed that the cell pellet at the bottom of the 68% Percoll fraction was maximally enriched with SYTO9-positive cells (78.22% ± 1.50%) and the least amount of PI-positive cells (2.97% ± 0.29%) (Fig. 2G, the right bottom panel in Fig. 2H–J). A ring of dead cells was observed at the top of the 48% fraction (Fig. 2G, first left top panel in Fig. 2H–J).

The CFU of each Percoll fraction showed the highest content of CFU-positive cells (72.02% ± 6.27%) in the 68% Percoll fraction, which was significantly higher than other fractions (Fig. 2K

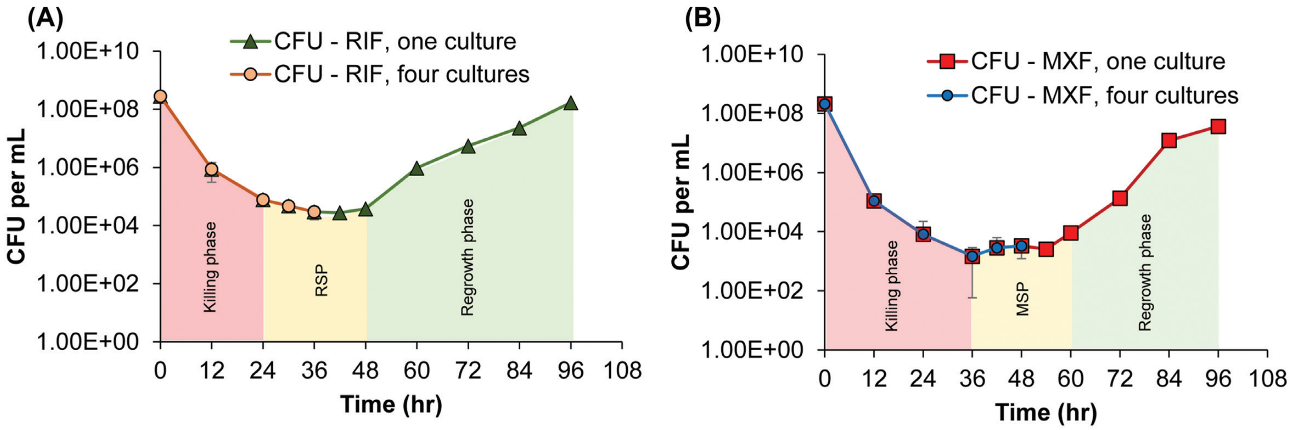


Figure 1. Susceptibility profile of the *Msm* cells exposed to rifampicin and moxifloxacin determined on antibiotic-free plates. **(A)** Rifampicin susceptibility profile of biological triplicate cultures (orange filled circles) up to 36 hr of rifampicin (RIF; 75 μg mL^{-1}; 1.75x MBC) exposure and of the fourth culture (green triangles) up to 96 hr. **(B)** Moxifloxacin susceptibility profile of biological triplicate cultures (blue filled circles) up to 48 hr of moxifloxacin (MXF; 0.5 μg mL^{-1}; 3.75x MBC) exposure and of the fourth culture (red squares) up to 96 hr. The three phases of response of *Msm* cells upon antibiotic exposure has been shown with coloured filled space with: pink for the killing phase (0–24 hr for rifampicin exposure and 0–36 hr for moxifloxacin exposure); yellow for the antibiotic surviving phase (24–48 hr for rifampicin surviving phase (RSP) and 36–60 hr for moxifloxacin surviving phase; MSP); and green for the regrowth phase (48–96 hr for rifampicin regrowth phase and 60–96 hr for moxifloxacin regrowth phase).

and L). All the other fractions contained negligible proportions of the CFU-positive cells insufficient for total RNA extraction (Fig. 2K and L). The live-cell enrichment, which was visible in the Figs. 2I and 2K, were not reflected visually in Fig. 2L as the CFU data were plotted in $\log_{10}$ scale. The SYTO9-positive cells were enriched to the same extent (~76%) in the repeat Percoll gradient centrifugations performed in the absence and presence of rifampicin (Fig. S7A, B, compare with C, D, respectively; compare these values with Fig. 2I, J and Fig. S5C and D, Supporting Information). Thus, the 68% Percoll fraction contained the maximally enriched CFU-cum-SYTO9-positive cells. The CFU proportions, in the presence and absence of rifampicin, were almost the same (Fig. S7E, Supporting Information). Thus, the presence or absence of bactericidal concentration of rifampicin did not affect the enrichment process using Percoll gradient centrifugation. Nonetheless, to maintain the metabolic status of the cells unaltered, the cells enriched by the Percoll gradient centrifugations performed in the presence of the antibiotic, were used for total RNA preparation.

Enrichment of CFU-cum-SYTO9-positive cells from the moxifloxacin-survivors

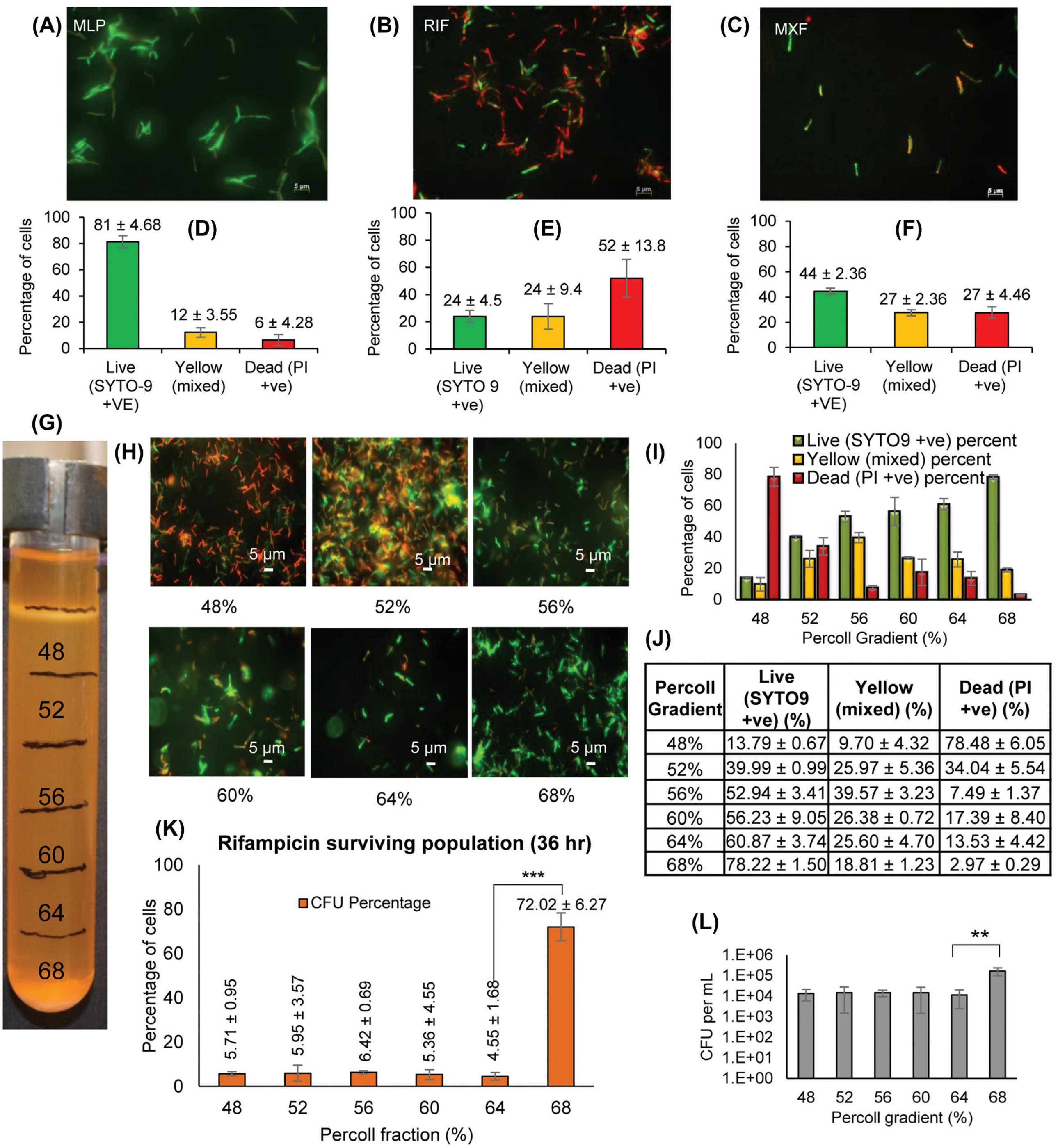
Different ranges of Percoll gradient were set up to get CFU-cum-SYTO9-positive enriched population from the 48-hr moxifloxacin-surviving population (Supplementary Text and Figs S8 and S10 in Supplementary Materials). The maximal enrichment of SYTO9-positive cells from the 48-hr moxifloxacin-surviving population could be achieved using the 40%–60% Percoll gradient containing 3.75x MBC moxifloxacin (Fig. 3A). The SYTO9/PI staining showed that the 60% Percoll fraction was 73.91% $\pm$ 6.20% enriched with SYTO9-positive cells, with only 0.27% $\pm$ 0.27% of PI-positive cells (Fig. 3B–D). The CFU of each Percoll fraction showed that the 60% Percoll fraction contained the highest content of CFU-positive cells (89.38% $\pm$ 2.19%), which was 1-to-2-$\log_{10}$-fold significantly higher than those in the other fractions (Fig. 3E and F). The proportion of the CFU-positive and SYTO9-positive cells (live cells) in each Percoll fraction confirmed that the 60% Percoll fraction was maximally enriched with the CFU-cum-SYTO9-positive cells surviving in the presence of moxifloxacin (Fig. 3D and E). These cells were used for the total RNA preparation.

Verification of the suitability of Percoll gradient fractionation

We verified the efficacy of the 48%–68% (rifampicin-containing) and 60% of the 40%–60% (moxifloxacin-containing) Percoll gradients for the enriched fractionation of live cells from the mid-log phase populations unexposed to the antibiotics. The 48%–68% and 40%–60% Percoll gradients of the mid-log phase populations also showed 93%–95% of SYTO9-positive cells in the 68% and 60% Percoll fractions of the gradients, respectively (Fig. S11A–D and S12A–D, Supporting Information). The CFU profile of the fractions showed ~4-$\log_{10}$-fold significantly higher number of cells (~99%) in the 68% and 60% fractions, as compared to the proportion of the cells in the other fractions (Fig. S11E and F and Fig. S12E and F, Supporting Information). The data from these control experiments confirmed the efficacy and suitability of the 48%–68% and 40%–60% Percoll gradients for the enriched fractionation of CFU-cum-SYTO9-positive cells from the 36 hr rifampicin-surviving and 48 hr moxifloxacin-surviving populations, respectively.

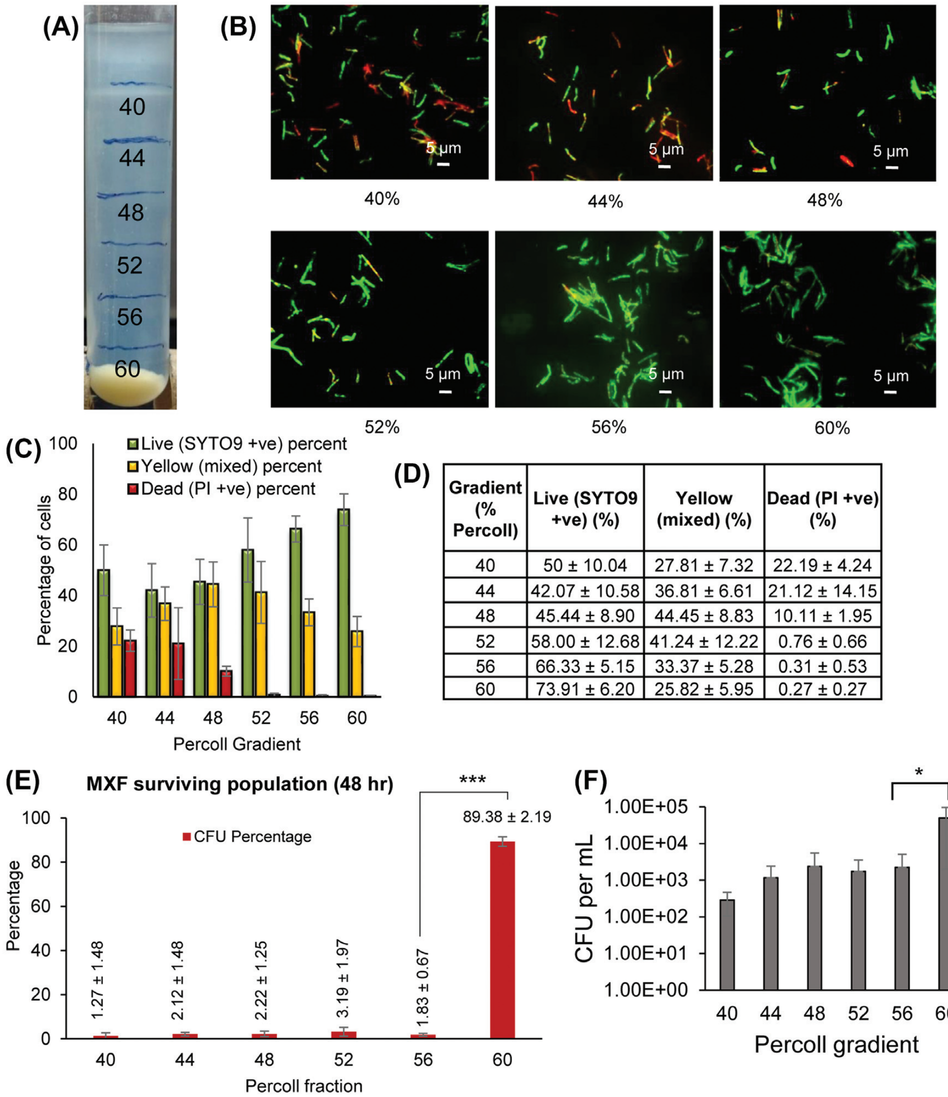
RNA profile of the CFU-cum-SYTO9-positive rifampicin/moxifloxacin-survivors

The DNase I treated total RNA samples extracted from the unfractionated 36 hr rifampicin-surviving population and the 48 hr moxifloxacin-surviving population, without the enrichment for CFU-cum-SYTO9-positive cells, were found to contain mostly degraded RNA (Fig. 4A and B, respectively). This was due to the presence of RNA from the large proportions of dead and damaged cells, as expected, despite the presence of SYTO9-positive cells. On the contrary, the DNase I treated total RNA samples extracted from the mid-log phase cells, and the CFU-cum-SYTO9-positive cells of the 68%/60% Percoll fractions of the 36-hr-rifampicin/48-hr-moxifloxacin-surviving populations showed intact 23S, 16S, and 4–5S RNA profiles (Fig. 4C–E, respectively). The bioanalyser based 23S/16S rRNA ratio and



Percoll Gradient	Live (SYTO9 +ve) (%)	Yellow (mixed) (%)	Dead (PI +ve) (%)
48%	13.79 ± 0.67	9.70 ± 4.32	78.48 ± 6.05
52%	39.99 ± 0.99	25.97 ± 5.36	34.04 ± 5.54
56%	52.94 ± 3.41	39.57 ± 3.23	7.49 ± 1.37
60%	56.23 ± 9.05	26.38 ± 0.72	17.39 ± 8.40
64%	60.87 ± 3.74	25.60 ± 4.70	13.53 ± 4.42
68%	78.22 ± 1.50	18.81 ± 1.23	2.97 ± 0.29

Figure 2. Representative images of the SYTO9-/PI-stained cells and the enrichment of the SYTO9-positive cells using Percoll gradient centrifugation. Images of the cells from: **(A)** the mid-log phase cultures, **(B)** the 36 hr rifampicin surviving population, and **(C)** the 48 hr moxifloxacin surviving population. **(D—F)** Bar graphs showing the percentage of SYTO9 positive (green bar), PI positive (red bar), and SYTO9 & PI positive (mixed) cells (yellow bar) in the: **(D)** mid-log phase cultures, **(E)** 36 hr rifampicin surviving population, and **(F)** 48 hr moxifloxacin surviving population, respectively ($n \geq 300$). **(G)** The 48%-68% Percoll gradient containing 75 µg mL^{-1} rifampicin, showing cells at 48% Percoll and a cell pellet at 68% Percoll. **(H)** The SYTO9-/PI-stained samples of the six Percoll fractions. **(I, J)** The percentage of SYTO9-positive, PI-positive, and SYTO9- and PI-positive (yellow) cells out of the total number of stained cells counted from each Percoll fraction ($n \geq 300$). **(K)** The proportions of CFU-positive cells (CFU percentage) in each Percoll fraction out of the respective total CFU from all the fractions ($n = 3$). **(L)** The CFU per mL of the cells present in each percentage of Percoll of the gradient in **(G)**. Statistical significance was calculated using one-way ANOVA. $^{**}P \leq 0.01$, $^{***}P \leq 0.001$.



Gradient (% Percoll)	Live (SYTO9 +ve) (%)	Yellow (mixed) (%)	Dead (PI +ve) (%)
40	50 ± 10.04	27.81 ± 7.32	22.19 ± 4.24
44	42.07 ± 10.58	36.81 ± 6.61	21.12 ± 14.15
48	45.44 ± 8.90	44.45 ± 8.83	10.11 ± 1.95
52	58.00 ± 12.68	41.24 ± 12.22	0.76 ± 0.66
56	66.33 ± 5.15	33.37 ± 5.28	0.31 ± 0.53
60	73.91 ± 6.20	25.82 ± 5.95	0.27 ± 0.27

Figure 3. Percoll gradient and percentage of SYTO9-/PI-positive cells in the 48 hr moxifloxacin surviving population. **(A)** 40%-60% Percoll gradient with 4.0% increment, in the presence of 0.5 $\mu g\ mL^{-1}$ moxifloxacin in the gradient, showing cells at top of 40% Percoll and a cell pellet at the 60% Percoll fraction. **(B)** SYTO9-/PI-stained cells in the six different Percoll fractions. **(C, D)** The percentage of SYTO9-/PI-positive cells out of the total number of stained cells counted from each Percoll fraction ($n \geq 300$). **(E)** The proportions of CFU-positive cells (CFU percentage) in each Percoll fraction out of the total CFU from all the fractions ($n = 3$). **(F)** The CFU per mL of the cells in each Percoll fraction of the gradient in **(A)**. Statistical significance was calculated using one-way ANOVA. $^{*}P \leq 0.1$, $^{***}P \leq 0.001$.

the RNA integrity number (RIN) for the CFU-cum-SYTO9-positive enriched cells were: 1.8 and 9.9 for the mid-log phase sample, 2.1 and 9.3 for the 36-hr rifampicin-surviving population, and 2.0 and 9.6 for the 48-hr moxifloxacin-surviving population (Fig. 4C–E, respectively). These values confirmed the high structural integrity of the total RNA (Schroeder *et al.* 2006). The Fig. S13 (Supporting Information) gives the flow chart of the protocol for the isolation of maximally enriched CFU-cum-SYTO9-positive *Msm* cells from the rifampicin/moxifloxacin-surviving populations.

Gene expression profiles of the rifampicin/moxifloxacin-surviving populations

The gene expression profile of the rifampicin- and moxifloxacin-surviving populations revealed statistically significant (>2-fold change; $P < 0.001$) upregulation of 1069 and 155 genes, and downregulation of 238 and 57 genes, respectively. A four-dimensional Venn-diagram (Heberle *et al.* 2015) summarises the differences and commonalities between the gene expression profiles showing 57 genes commonly upregulated and 26 genes commonly downregulated (Fig. 5A; Excel Table S1, Supporting Information). The genes were categorised based on their function (Baloni *et al.* 2014) and counted for the number of genes upregulated and downregulated along with percentage (>2-fold change; $P < 0.001$) for each functional category (Fig. 5B; Excel Table S1, Supporting Information). The percentage shows the proportion of the genes in the > 2-fold change category ($P < 0.001$) out of the total number of genes up/downregulated in the functional category of $P < 0.05$ (Fig. 5B; Excel Table S1, Supporting Information). The number and percentage of the genes up/downregulated in the rifampicin-surviving population were higher than in the moxifloxacin-surviving population. Among the energy metabolism and biosynthetic pathways, most of the

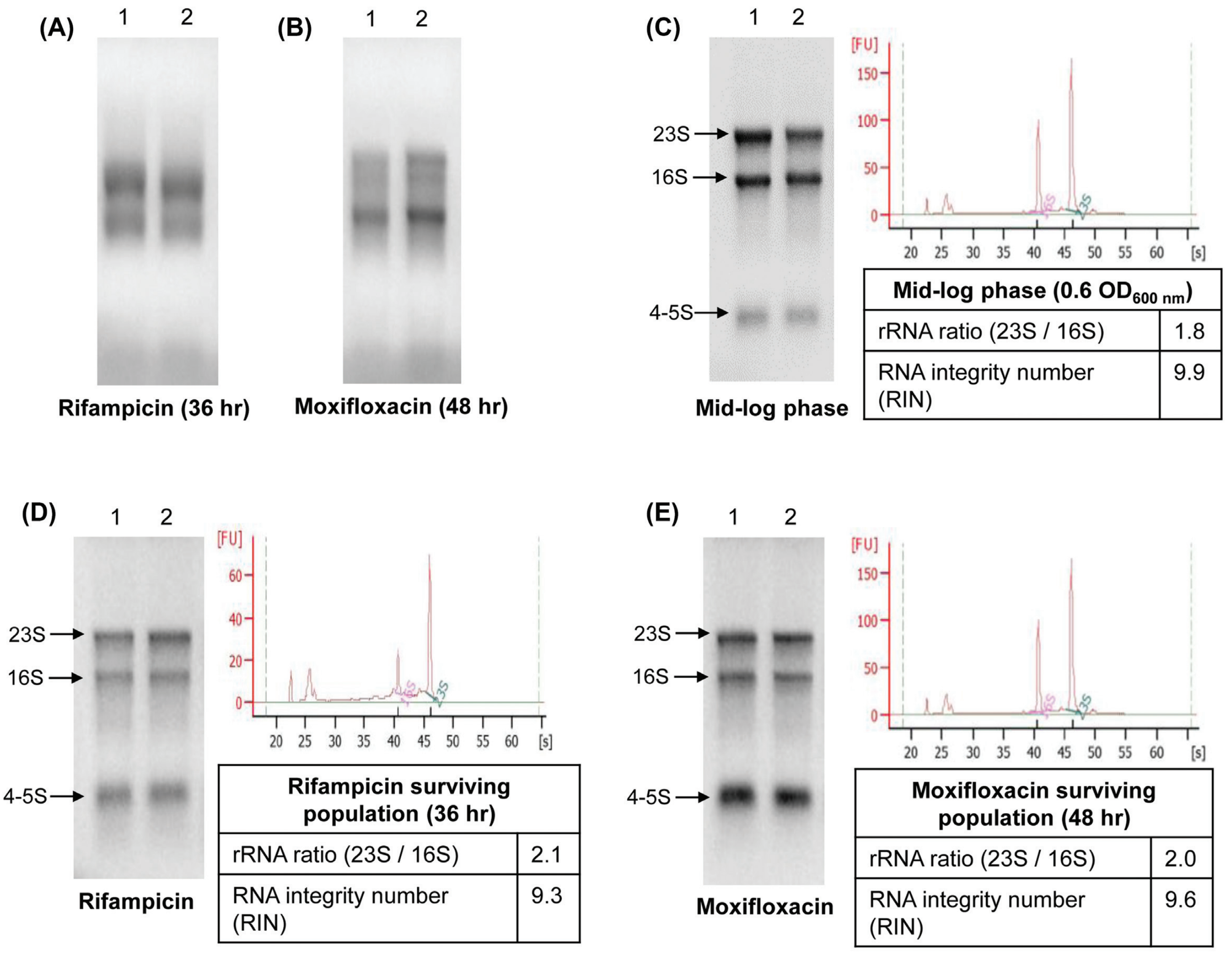
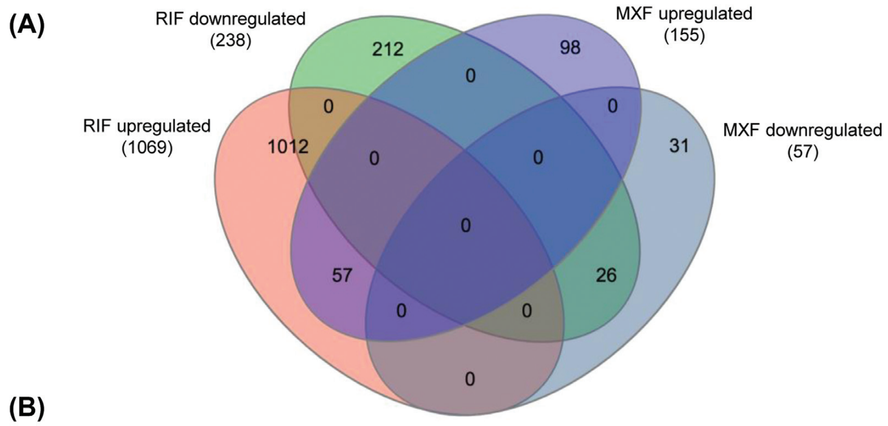


Figure 4. Total RNA profile on 1% denaturing agarose gel and the values determined using Bioanalyser. **(A–E)** Profile of the total RNA extracted from the replicate samples of the whole unfractionated population surviving against: **(A)** rifampicin (36 hr exposure) and **(B)** moxifloxacin (48 hr exposure). The total RNA profile of: **(C)** the mid-log phase cells; and of the SYTO9-positive live enriched cells in the: **(D)** 68% Percoll fraction from the 36 hr rifampicin surviving population and **(E)** 60% Percoll fraction from the 48 hr moxifloxacin surviving population. Lanes 1 & 2 show the RNA integrity profile of the total RNA samples extracted from biological replicates. The rRNA ratio and RNA integrity number (RIN) were from the bioanalyser data.

genes pertaining to respiration and electron transport were significantly upregulated while ribosomal proteins and glycolysis enzymes were significantly downregulated (Fig. 5C–F). The differential regulation of DNA repair genes in the rifampicin- and moxifloxacin-surviving populations showed the bacterial strategy to overcome the stress-induced damages through mutagenesis (Swaminath *et al.* 2020a, Jakkala *et al.* 2020; Sebastian *et al.* 2017), and prepare for growth in the presence of antibiotics (Fig. 5G).

Assessment of the reliability of the method

From the microarray-based gene expression profile of the tubercle bacilli surviving in the sputum of 17 tuberculosis patients undergoing treatment with isoniazid, rifampicin, pyrazinamide, and ethambutol for 14 days, Walter and co-workers identified 82 genes, which confer antibiotic tolerance (Walter *et al.* 2015). We could find 42 *Msm* orthologue genes (Kapopoulou, Lew and Cole 2011) from among the 82 genes of *Mtb*. The Microarray raw data of the *Msm* 36-hr/48-hr rifampicin/moxifloxacin-surviving populations (Excel Table S1, Supporting Information) were compared with the microarray data of the corresponding *Mtb* orthologue genes reported (Walter *et al.* 2015). Of the 42 *Msm* orthologues, the microarray-based altered expression levels of altogether 21 orthologue genes of the rifampicin- and moxifloxacin-surviving *Msm* populations matched with significance with those in the *Mtb* profile (Fig. S14, Supporting Information). Of the 21 genes, 5 genes were commonly upregulated, and 11 genes were commonly downregulated, in both the rifampicin- and moxifloxacin-surviving populations and in the *Mtb* sample (Table 1). The comparison of the microarray raw data of the *Mtb* and *Msm* orthologue genes for antibiotic tolerance, with statistical significance, are given in the Excel Table S2 (Supporting Information). Individually, 19 orthologue genes of the rifampicin-surviving population and 18 orthologue genes of the moxifloxacin-surviving population matched with the gene expression profile of the *Mtb* sample (Excel Table S2, Supporting Information). This hinted that the rifampicin/moxifloxacin-surviving *Msm* populations and the *Mtb* population from the antibiotic-treated patients were probably of comparable metabolic status. This statistically significant comparability validated the reliability and utility of the method to enrich and isolate antibiotic tolerant *Msm* populations surviving in the presence of bactericidal concentrations of rifampicin/moxifloxacin.



(B)

Function	Number of genes$ (%)				
		Rifampicin (36 hr)		Moxifloxacin (48 hr)	
	Microarray#	↑	↓	↑	↓
Cell wall and cell processes	874	117 (26)	25 (7)	21 (15)	5 (3)
Conserved hypothetical	1816	322 (32)	56 (18	40 (5)	12 (4)
Information pathways	848	147 (33)	47 (23)	28 (8)	7 (4)
Intermediary metabolism and respiration	1645	235 (29)	64 (19)	34 (5)	21 (6)
Lipid metabolism	446	69 (30)	10 (10)	10 (5)	6 (7)
PE-PPE families	10	2(40)	1 (50)	0 (0)	0 (0)
Regulatory proteins	193	26 (22)	8 (22)	5 (5)	2 (7)
Virulence, detoxification and adaptation	363	62 (29)	8 (18)	8 (4)	2 (4)
Unknown	341	40 (25)	16 (23)	3 (2)	1 (1)

All the genes are not present on the array; $ ≧2-fold, Significance at p<0.001

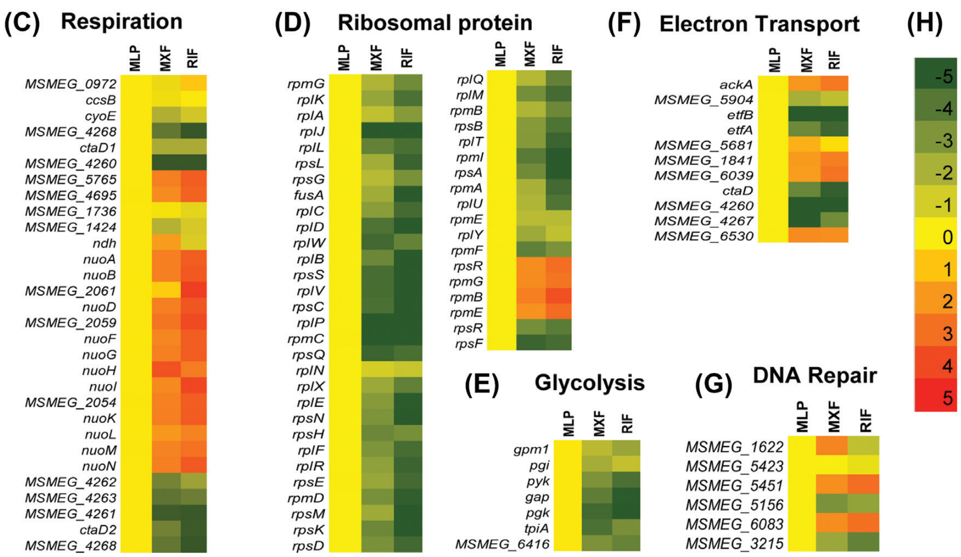


Figure 5. Microarray data analysis of the rifampicin- and moxifloxacin-surviving *Msm* populations. **(A)** Four-dimensional Venn-diagram comparing the genes upregulated and downregulated at ≧2-fold, with significance at $P < 0.001$. **(B)** Summary of the genes upregulated and downregulated at ≧2-fold, with significance at $P < 0.001$, categorised according to function. **(C-G)** Heat map of energy metabolism, biosynthetic pathways and repair genes in the rifampicin- and moxifloxacin-surviving populations, as compared with that of the mid-log phase (MLP) cells, at $P < 0.05$. **(H)** The 3 colour-scale for the heat map showing gradients for each fold change calculated at log base 2. RIF, rifampicin; MXF, moxifloxacin.

DISCUSSION

Merits of Percoll gradient fractionation of cells in the presence of antibiotic

In the present method, the rifampicin-/moxifloxacin-exposed *Msm* cells surviving in the presence of the respective antibiotic were enriched, isolated, and validated. This would have maintained the gene expression profile of the antibiotic-surviving populations throughout the procedure. The statistically significant correlation of the *Msm* data with the expression profile of 21 of the 42 antibiotic tolerance associated genes of *Mtb* from the sputum of 17 tuberculosis patients undergoing treatment (Walter *et al.* 2015), was an indication of this fact (Excel Table S2, Supporting Information). Further, we have successfully standardised Percoll gradient fractionation for the isolation of CFU-cum-SYTO9-positive enriched population of the

Table 1. Comparison of the expression levels of antibiotic tolerance genes of *Msm* with those of *Mtb*.

Identifier (*Mtb*)	*Msm* orthologue	Gene name	*Msm*—RIF	*Msm*—MXF	*Mtb*—day 14 Walter *et al.* (2015)	
		Upregulated				Cluster#
Rv1931c	MSMEG_1106		2.99 ± 0.08	2.25 ± 0.40	6.9	**A**
Rv0221	MSMEG_0290		2.74 ± 0.02	1.92 ± 0.40	11.6	**B**
Rv1273c	MSMEG_5008		2.07 ± 0.07	1.57 ± 0.42	4.5	
Rv1814	MSMEG_3666		2.73 ± 0.08	1.99 ± 0.40	3.4	
Rv0753c	MSMEG_1498	*mmsA*	3.52 ± 0.49	1.99 ± 0.42	3.4	**C**
		Downregulated				
Rv0640	MSMEG_1346		− 4.12 ± 0.26	− 2.46 ± 0.20	− 2.1	**D**
Rv2245	MSMEG_4327		− 4.94 ± 0.20	− 3.56 ± 0.70	− 3.0	
Rv3197	MSMEG_1954		− 0.56 ± 0.00	− 1.40 ± 0.40	− 1.8	
Rv0704	MSMEG_1439	*rplB*	− 5.37 ± 0.77	− 3.46 ± 0.44	− 8.5	**E**
Rv2031c	MSMEG_3932		− 3.70 ± 0.13	− 4.53 ± 0.50	− 21.6	
Rv2185c	MSMEG_4251		− 1.70 ± 0.19	− 2.67 ± 0.75	− 16.7	
Rv2455c	MSMEG_4646		− 2.22 ± 0.07	− 1.64 ± 0.39	− 5.7	
Rv2460c	MSMEG_4672	*clpP*	− 5.23 ± 0.08	− 4.59 ± 0.33	− 4.3	
Rv3583c	MSMEG_6077		− 4.13 ± 0.43	− 4.20 ± 0.86	− 8.7	
Rv3874	MSMEG_0065		− 3.73 ± 0.04	− 4.53 ± 0.40	− 7.6	
Rv3875	MSMEG_0066		− 4.20 ± 0.67	− 4.95 ± 0.31	− 5.8	

The comparison of the expression levels of the genes involved in antibiotic tolerance in the tubercle bacilli from the day 14 sputum samples of the 17 patients undergoing treatment with isoniazid, rifampicin, pyrazinamide and ethambutol (Walter *et al.* 2015), with their respective *Msm* orthologues in the 36 hr *Msm* rifampicin-surviving cells (*Msm*-RIF) and 48 hr *Msm* moxifloxacin-surviving cells (*Msm*-MXF). The expression in the mid-log phase cells was kept as the normalisation control (n = 2). The genes were grouped into different clusters based on the progressive change in the expression with increasing duration of treatment (antibiotic tolerance specific genes) (Walter *et al.* 2015). #Cluster **A**, **B** = demonstrated little to no change between days 0 and 2, but significant induction by days 7 and 14. #Cluster **C**, **D** = strongly induced on day 2 of the antibiotic exposure. #Cluster **E** = significantly downregulated at days 7 and 14. The data for the median fold-change in the expression was for *Mtb* day 14, relative to day 0 (Walter *et al.* 2015). Expression fold-change for *Msm*-RIF and *Msm*-MXF are the mean value with SD values (n = 2). $P \leq 0.05$, was considered as cut-off for change in the expression of the genes in microarray data.

rifampicin-/moxifloxacin-survivors of *Mtb* cells as well (unpublished observations). Therefore, it would be possible to isolate antibiotic-survivors of other mycobacterial species and of other bacterial systems against different antibiotics using specific ranges of Percoll gradients. Further, it would also be possible to isolate specific antibiotic-survivor phenotypes from the enriched majority of the survivor population.

Shortcomings of the method

One of the shortcomings of the method is that it was practically impossible to get the total population of antibiotic-survivors enriched. A total of ∼28% and ∼11% of the CFU-cum-SYTO9-positive cells from the rifampicin- and moxifloxacin-surviving populations, respectively, got distributed into other fractions of the respective Percoll gradient, albeit in small proportions. This distribution hinted at the differences in their buoyant densities, which would be indicative of differences in their metabolic status. Nevertheless, these cells would still be useful for studies at the single-cell level. Second, the possibility of some specific types of antibiotic survivors getting fractionated into the Percoll fractions containing minor proportions of CFU-cum-SYTO9-positive cells could not be ruled out. Third, none of the Percoll fractions of either of the antibiotic-treated cultures showed enrichment of yellow-fluorescing viable/non-viable cells. Hence, they could not be isolated even though they would contain certain proportion of live cells.

The relevance of CFU-cum-SYTO9-positive readout for live cell quantitation

SYTO9 PI staining is a well-known staining technique for the easy identification of intact/damaged cells. Although staining *Mycobacterium tuberculosis* with PI for flow cytometry has limitations (Soejima *et al.* 2009), PI-stained *M. smegmatis* cells have been used for fluorescence microscopy (Emani *et al.* 2013; Jha *et al.* 2020; Ravindran *et al.* 2020; Ko and Hatfull 2020). Nevertheless, the combined CFU-cum-SYTO9-positive readout as the criterion for determining the enrichment of live, viable cells overcomes any limitation due to SYTO9 staining. The highest proportions of the CFU-cum-SYTO9-positive cells from both the mid-log phase cultures (∼99%) and the rifampicin/moxifloxacin-surviving populations (72%/89%) were found in the 68%/60% Percoll fractions, respectively (compare Figs 2L and 3F with Figs S11F and S12F (Supporting Information) respectively). This confirmed that the rifampicin/moxifloxacin-survivors in the 68%/60% fractions were truly the CFU-cum-SYTO9-positive cells enriched from the respective antibiotic-surviving population.

Gene expression features of the rifampicin/moxifloxacin-surviving populations

The transcriptome data gave important insights into the bacterial response to antibiotic stress. The change in the gene expression profiles of the rifampicin- and moxifloxacin-surviving populations revealed a strategy of countering the effect of antibiotics through different metabolic changes and the survival strategies thereof (Stokes *et al.* 2019). Most of the cell wall transport proteins such as ABC transporters, amino acid transporters, and others were upregulated in both the surviving populations. It showed that the cells were actively involved in transporting substrates into the cells for survival (He and De Buck 2010; Cassio Barreto de Oliveira and Balan 2020). The genes like mycolic acid synthase (*umaA*, *MSMEG_0913*) and *murA* (*MSMEG_4932*), which are reported as potential targets for antituberculosis drugs (He and De Buck 2010; Xu *et al.* 2014), and many other genes involved in lipid metabolism, synthesis of cell wall components, information pathways and ribosomal proteins, were downregulated. It showed the bacterial survival strategy of slowing down growth in the presence of bactericidal concentration of antibiotics (Tuomanen *et al.* 1986; Keren *et al.* 2011; Gomez *et al.* 2017). The transcriptional regulatory genes involved in the synthesis of TetR family proteins, which are reported for their role in multidrug resistance (Ramos *et al.* 2005), were found to be upregulated, which also showed bacterial survival strategy against antibiotics.

Glycerol kinase (*glpK*; *MSMEG_6229*) which is involved in glycerol metabolism, is associated with drug resistance in clinical isolates (Bellerose *et al.* 2019). The *glpK* gene increases the sensitivity of mycobacteria to rifampicin and moxifloxacin in the presence of glycerol as the carbon source. Similarly, changes in glycerol catabolism influence the sensitivity of bacteria to antibiotics and survival (Bellerose *et al.* 2019). This justifies the reason for the downregulation of *glpK* and glycolysis-related genes, in both the rifampicin- and moxifloxacin-surviving populations. The virulence related genes in *Msm*, which is a saprophyte-cum-opportunistic-pathogen (Wallace *et al.* 1988; Pierre-Audigier *et al.* 1997; Alqurashi *et al.* 2019), is the indicator of its resemblance with the pathogenic species, *Mtb*. The Mce (mammalian cell entry) family proteins, which are on the mycobacterial cell wall, help the entry of the bacterium into macrophages and in the subsequent survival (Zaychikova and Danilenko 2020). The upregulation of the genes of the Mce family proteins in the rifampicin- and moxifloxacin-surviving populations of *Msm* is probably an indication of their role in some metabolic processes besides facilitating entry into macrophages. The commonality in the expression of the Mce proteins in the antibiotic exposed *Msm* and *Mtb* validates the use of *Msm* as a model organism to study the survival strategy of the pathogenic *Mtb*. The statistically significant comparability between the expression profiles of the 21 out of the 42 orthologue genes of the *in vitro* antibiotic exposed *Msm* and the *in vivo* antibiotic exposed *Mtb* further strengthens the suitability of *Msm* as a model system. Further, the comparative transcriptome profiles of the *Msm* to antibiotics with different modes of action showed the adaptability of the bacterium, although comparable changes in the expression levels were found for most of the genes.

SUPPLEMENTARY DATA

Supplementary data are available at *FEMSLE* online.

AUTHOR CONTRIBUTION

PA and AP designed the experiments; AP, SS and KJ performed the experiments; PA, AP, KJ and SS analysed the data; PA provided the materials and reagents for the experiments; PA supervised the experimental methods and analysis of the data. AP and PA wrote the manuscript. All the authors have read and approved the manuscript.

ACKNOWLEDGEMENTS

With highest respects and regards, PA dedicates this work as a tribute to Prof. Thekkepat Ramakrishnan (late), who led the pioneering, fundamental and foundation laying work on the biochemistry and molecular biology of *Mycobacterium tuberculosis* at Indian Institute of Science, Bangalore, India.

FUNDING

The work was supported in part by the DBT-IISc Partnership Programme (2012–2020), and by the infrastructure facilities provided by the DBT-IISc Partnership Programme (2012–2020), DST-FIST, UGC-CAS, ICMR-CAS, and Indian Institute of Science, in the MCB Dep't. AP, SS received Senior Research Fellowships from the CSIR, Government of India. KJ was a Senior Research Fellow of UGC, Government of India, and later Senior Research Fellow in a DBT project. SS, AP, and KJ were Research Associates of Indian Institute of Science, Bangalore. PA is Emeritus Scientist of the Indian Council of Medical Research (ICMR), Government of India.

Conflicts of interests. The content of this publication does not represent the official views of any of the funding agencies that supported this work. The authors declare no conflict of interest.

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